Ruminant Disease

Diseases in Neonatal Ruminants



Neonatal Morbidity and Mortality

Diseases of the new born calf and neonatal calf mortality are the major causes of economic losses in livestock production.

- Approximately 66% of all morbidity in the first 6 months and of all mortality in the first year of life occurs in the first month of life.



Figure 1: Age (weeks) factor in neonatal calf mortality.

Neonatal Morbidity and Mortality

- Management in neonatal period
  - acidosis
  - colostrum management
- Post-neonatal infection
  - enteric
  - respiratory
- others

Acidosis

- Respiratory acidosis
- Metabolic acidosis

Acidosis

- Respiratory acidosis: newborn stage
  - asphyxia (atelectasis):
    - caused by dystocia
    - insufficient pulmonary surfactant
  - CO_2 build up in the blood

$$CO_2 + H_2O \rightleftharpoons H_2CO_3 \rightleftharpoons H^+ + HCO_3^-$$

Acidosis

- Metabolic acidosis: 1–4^{+} weeks old
  - diarrhea:
    - HCO_3^- loss
    - abnormal fermentation: lactate accumulation
  - hypoxia: glycogen → lactate
  - anorexia: dysbiosis, abnormal fermentation

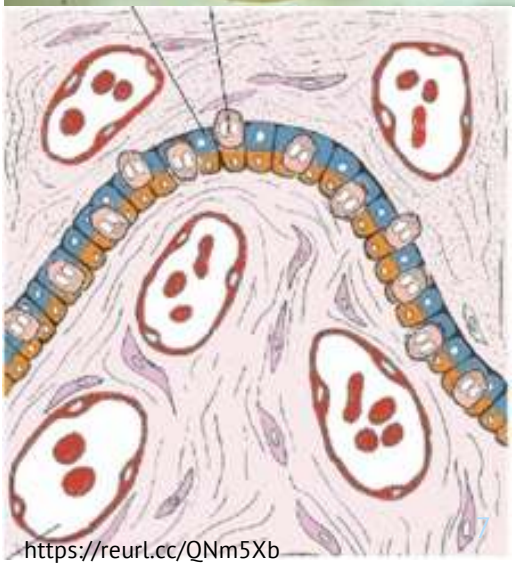
Acidosis

- Signs:
  - depressed, lethargy
  - cannot stand
  - loss of reflexes
  - hypothermia of extremities
  - coma

Neonate's immune system

- agammaglobulinaemia
  - syneitheliachorial placenta (結締組織絨毛膜胎盤): protein are unable to penetrate
- Maternal antibody: colostrum





https://reurl.cc/QNm5Xb

Neonate's Immune System

- The milk from first milking after parturition, that is the colostrum, has the highest IgG concentration.
- The concentration of the IgG in milk falls dramatically in the following milking

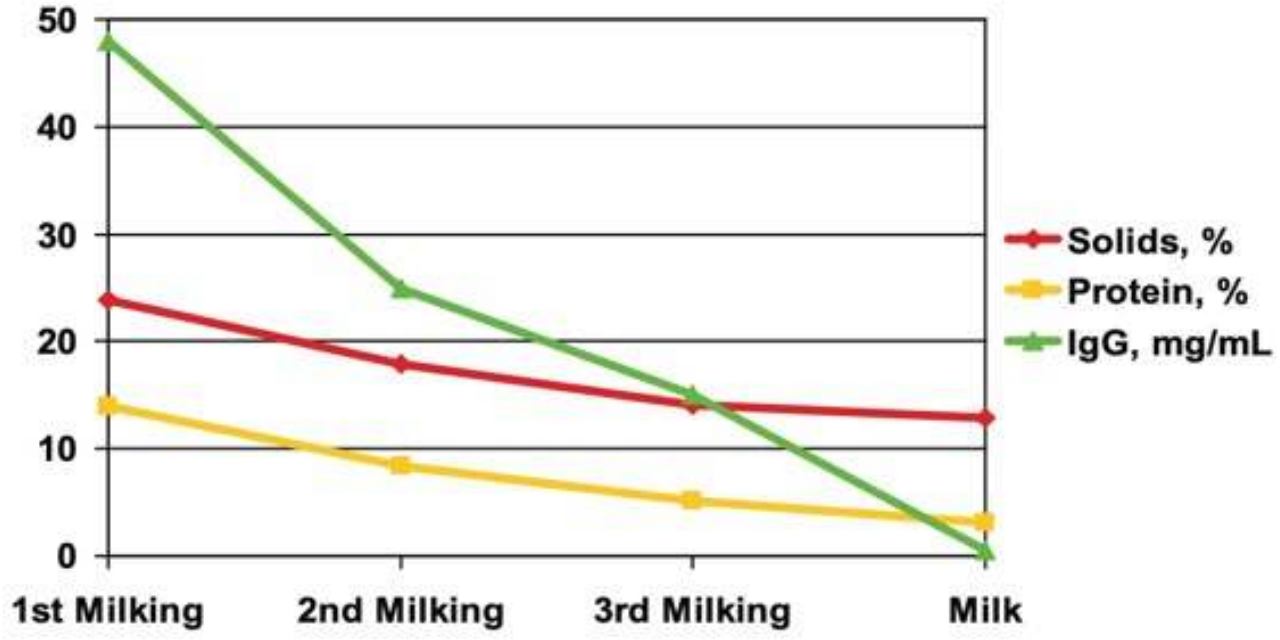
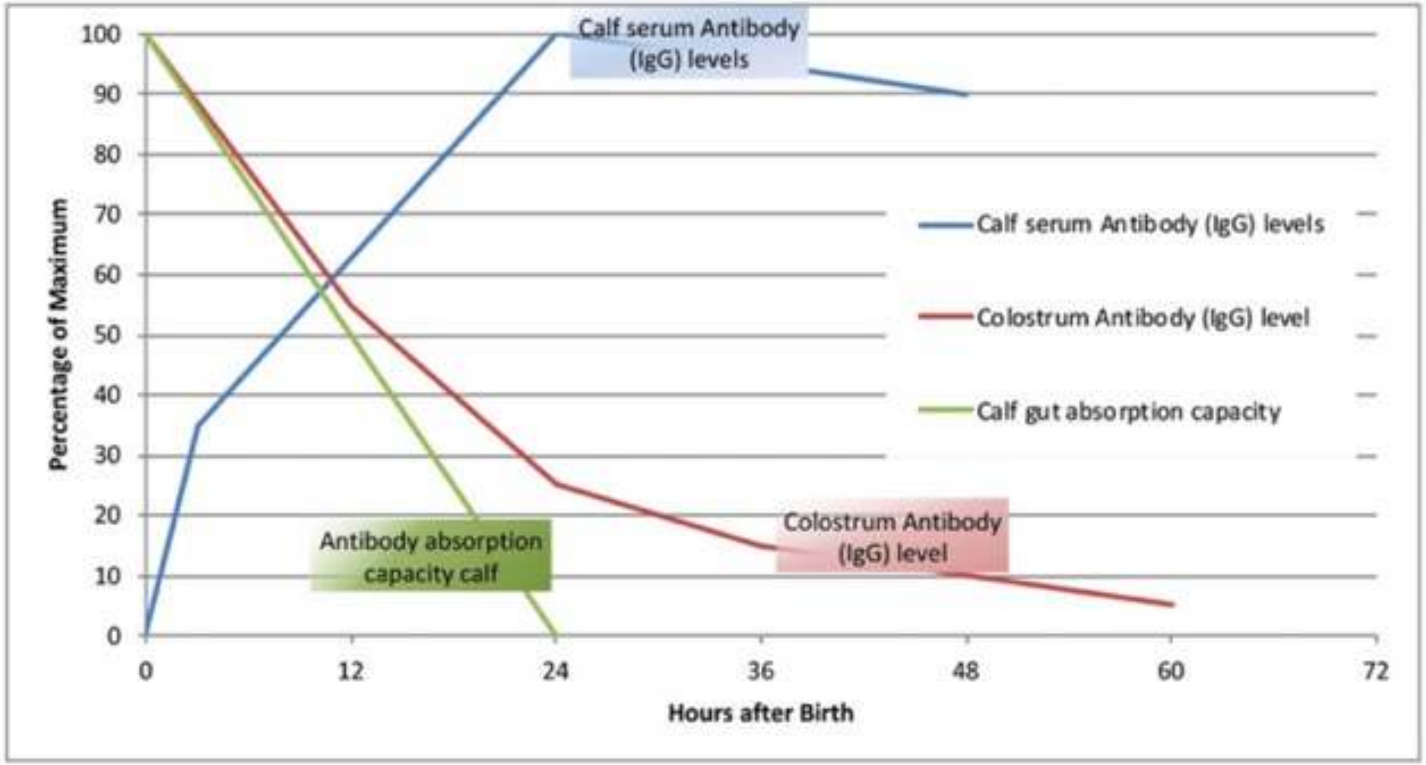


Fig. Typical composition of colostrum and transitional milk.
(Foley and Otterby, 1978)

Neonate's Immune System

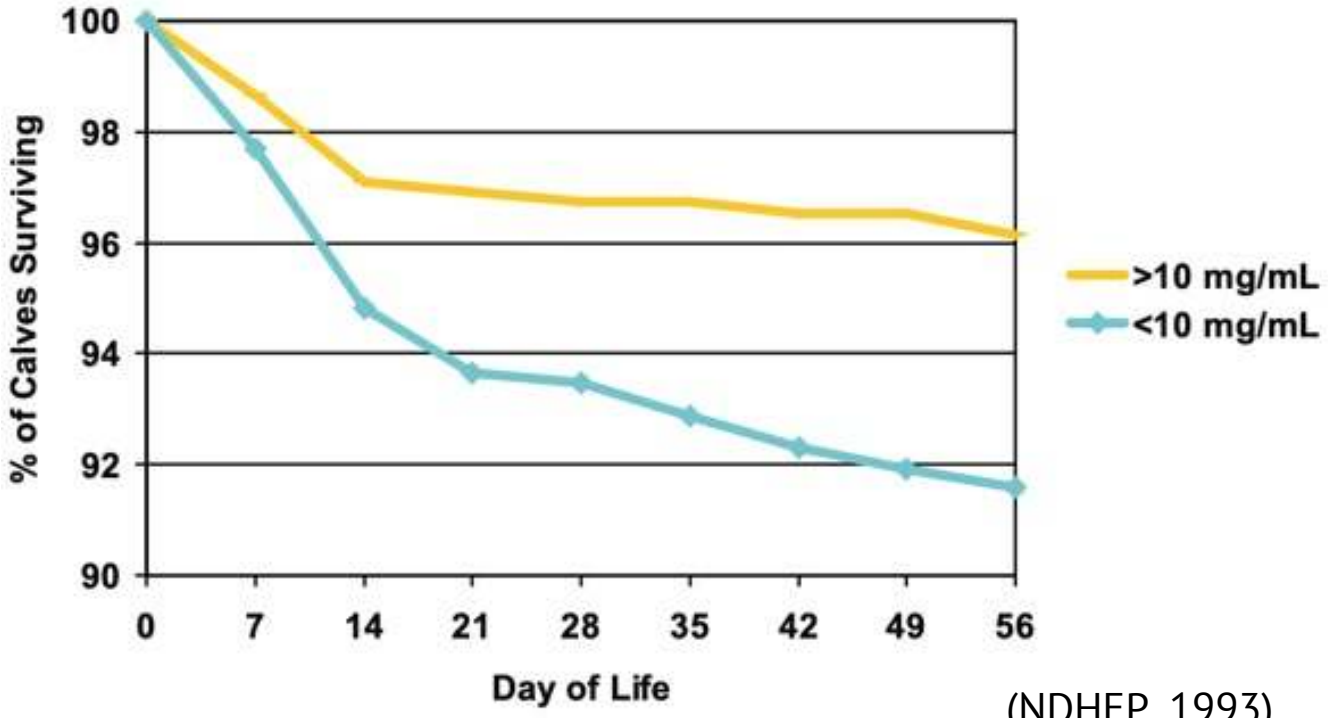


This clearly illustrates that the 'golden period' for colostrum administration is in those first 6 to 12 hours after birth. It is literally a race against time to get adequate levels of antibodies into the calf and if you do not succeed in those first 24 hours of a calf's life, that opportunity is lost.

https://anexa.co.nz/colostrum-calf-immunity-why-is-it-such-a-race-against-the-clock/

Neonate's Immune System

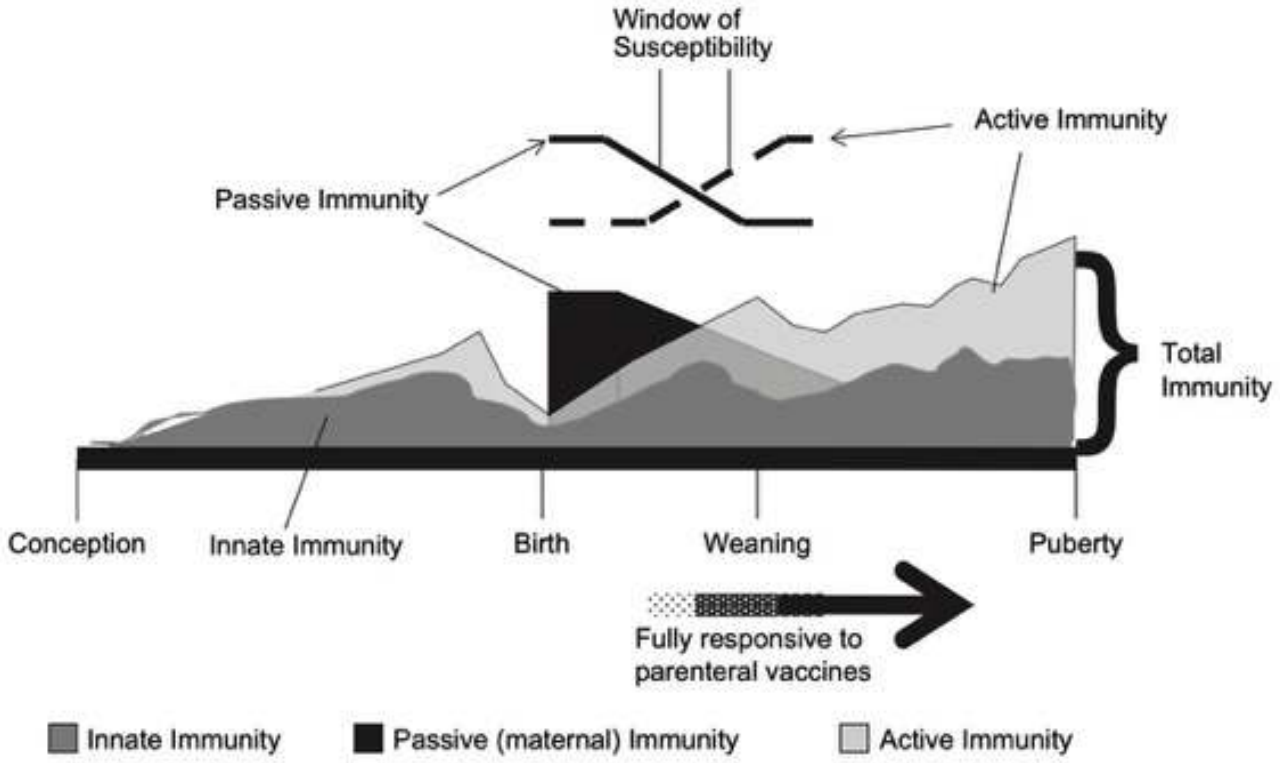
- Serum IgG concentration of a calf is > 10 mg/dL within 48 hours after birth: good passive immunity.



(NDHEP, 1993)

Immaturity of Neonate's Immune System

- The immune system continues to mature until approximately 6 months of age.
- The colostrum provides neonates with immunologic protection during at least the first 2–4 weeks of life.



(Chase *et al.*, 2008)

Colostrum Uptake for optimal passive immunity

- Feeding 7.5% of BW liters withing 4 hours of birth
  - or colostrum contained minimum 150 g IgG
- Boosting 5% of BW colostrum within 12 hours of birth
  - or further 7.5% of BW
    - ☞ Feeding colostrum 10–12% BW as a single initial feed
      - → Adequate passive immunity
      - → Moderate depression and discomfort in the calf

Neonatal Morbidity and Mortality

- Management in neonatal period
  - respiratory acidosis
  - colostrum management
- Post-neonatal infection
  - enteric
  - respiratory
- others

Neonatal Morbidity and Mortality

Table 1. Main Causes of Death in Dairy Calves within the First Month of Life (n = 498).

Cause of Death	n	%	Mean Age ± SD (days)
Chronic diarrhea	211	42.4	20.3 ± 5.6
Acute diarrhea	126	25.3	10.8 ± 5.2
Septicemia	54	10.8	7.7 ± 5.0
Cachexia/wasting syndrome	27	5.4	25.0 ± 3.7
Other causes	80	16.1	17.5 ± 7.8

Differences in age among diagnostic categories were assessed using the Kruskal–Wallis test ($H = 226.39$, $df = 4$, $p < 0.001$).

(Bednarski and Kupczynski, 2026)

Neonatal Morbidity and Mortality

- Most common causes leading to mortality in calves:
  - gastrointestinal infections
  - respiratory infections
  - umbilical/joint infections
  - abomasal/torsion disorders
  - congenital defects

Gastrointestinal infection

- Enteritis and septicemia:
  - Signs: diarrhea
  - Morbidity: 20–40%

    Mortality: 5–10 %



(Mee, 2023)



Gastrointestinal infection

- Risk factors:
  - failure of passive transfer
  - low plane of nutrition
  - contaminated milk or bedding
  - co-morbidity with respiratory system

Gastrointestinal infection

- Sequela:
  - death
    (malnutrition, dehydration, electrolyte imbalance, acidosis)
  - reduced average daily gain
  - older age at first calving
  - lower milk production

Gastrointestinal infection

- Bacteria:
  Escherichia coli, *Salmonella* spp., *Klebsiella* spp., *Clostridium* spp.
- Virus:
  rotavirus, coronavirus, bovine viral diarrheal virus, adenovirus, herpesvirus
- *Coccidia* spp., *Cryptosporidium* spp., other parasites

Enterotoxigenic *E. coli*

- 3-day-old calves
- White to yellow diarrhea
- Without blood/mucus
- Rapid dehydration, recumbency, electrolyte imbalance



Enterotoxigenic *E. coli*

- Adhere to the mucosa and proliferate in the lumen of intestine
  - enterotoxin: stimulate excess secretion of fluid form intestinal mucosa
  - bacterial arthritis, polyserositis, meningitis, pyelonephritis

Salmonellosis

- 2–6 weeks old calves
- *Salmonella* Dublin, *Salmonella* Typhimurium
- Foul-smelling diarrhea, often contains mucus, fibrin, or blood
- Rapid dehydration, electrolyte imbalance, weakness
- Secondary infection: pneumonia, arthritis, meningoencephalitis

Cryptosporidiosis

- 1–4 weeks old calves
- White to yellow diarrhea with odor smell
- Sometimes with blood in diarrheal feces
- Depression, anorexia, dehydration



Treating and Management in GI infection

Maintaining hydration and energy balance

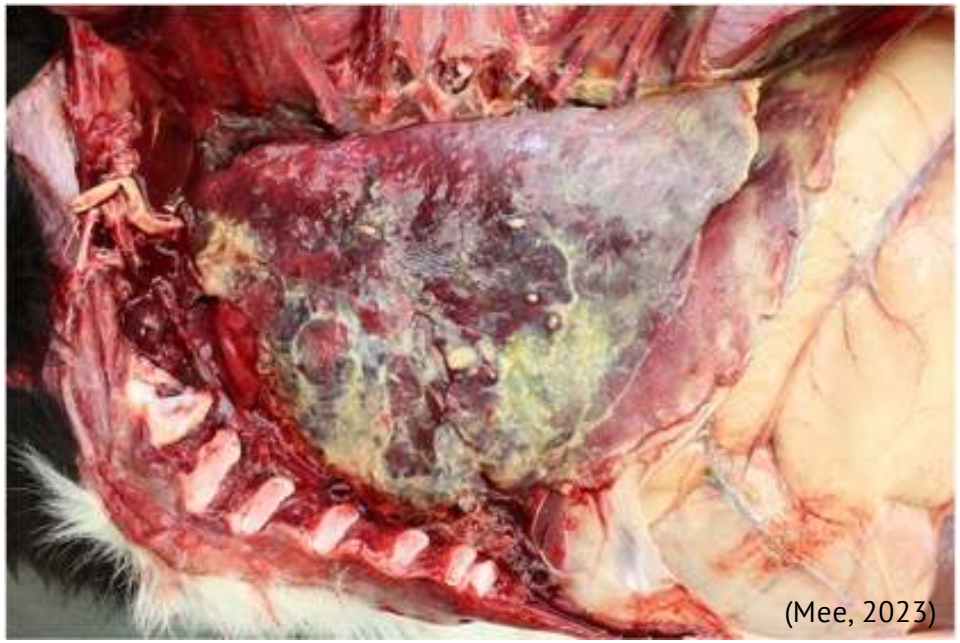
- Moderate to severe cases: withhold milk for 24 hours
- Proving electrolytes and energy: LR + glucose 50 g, oral, twice (at least)
- Enrofloxacin (2.5 mg/kg, IM, sid, for 3 days)
- IV infusion: LR (1,000 mL), glucose 50 g, methionine 20 mL

Respiratory infection

- Signs: nasal discharge, cough, pneumonia
  - Morbidity: 5–30%

    Mortality: 5–10 %

    (20–25% of mortality in pre-weaned)



(Mee, 2023)

Respiratory infection

- Risk factors:
  - failure of passive transfer
  - Intensive / mixed-age rearing
  - environment / housing design
  - co-infection/co-morbidity
- Sequela: negative effect on subsequent health, growth, reproduction, longevity, milk production and survival

Respiratory infection

- Bacteria:
  Mannheimia haemolytica, *Mycoplasma bovis*, *Pasteurella multocida*
- Virus:
  respiratory syncytial virus, coronavirus, bovine herpesvirus type 1, parainfluenza virus type 3, bovine viral diarrhea virus





Umbilical / joint infections

- Signs:
  - swelling, pain, heat, purulent discharge or delay in drying up of the umbilicus
  - occult infections
- Concurrent purulent arthritis

Umbilical / joint infections





Umbilical / joint infections

- Risk factors:
  - assisted calving
  - poor hygiene in the calving area
  - failure of passive transfer
  - improper umbilical care
  - inherent predisposition

Umbilical / joint infections

- Sequela:
  umbilical hernia, stunted growth, diarrhea, septicemia, septic polyarthritis, hepatic abscessation, peritonitis
- Environmental pathogens:
  E. coli, *Trueperella pyogenes*

Umbilical / joint infections

- Risk factors:
  - assisted calving
  - poor hygiene in the calving area
  - failure of passive transfer
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  - inherent predisposition

- Prevention:
  - calving in and maintaining a hygienic calving environment
  - moving the calf to a clean pen ASAP
  - umbilical care and regular inspection to detect early signs
  - good colostrum feeding management

Umbilical / joint infections

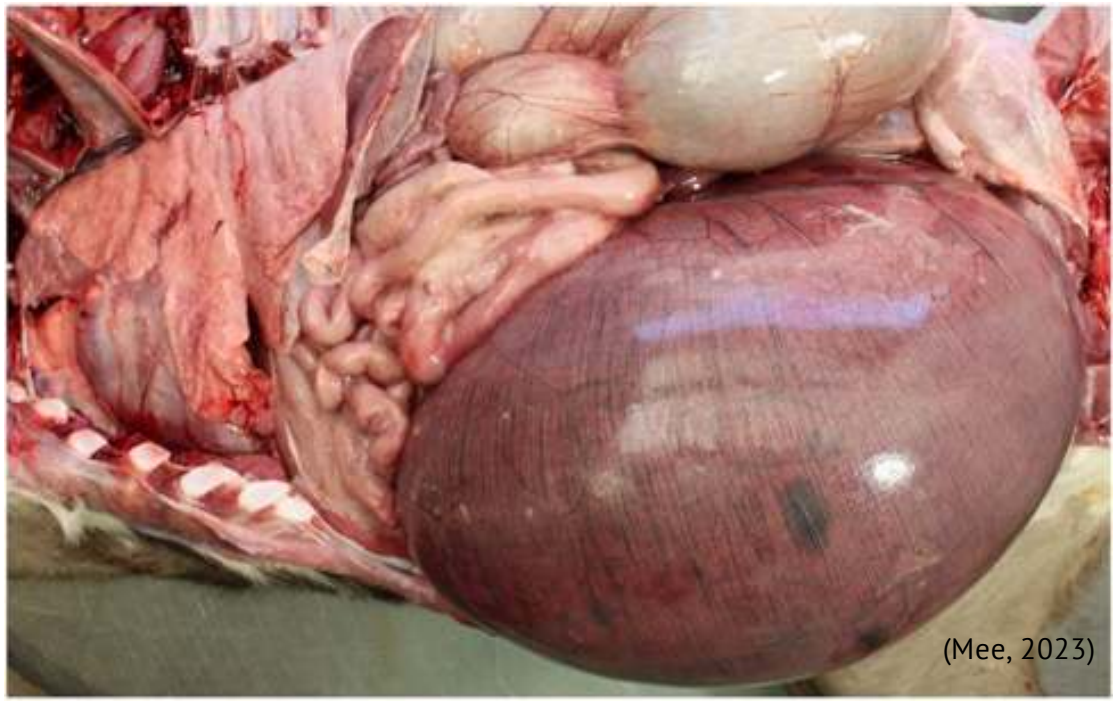
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https://livestock.extension.wisc.edu/articles/navel-care-for-beef-x-dairy-calves/

Abomasal syndrome

- Clinical signs associated with abomasal disorders, including abomasitis, tympany, ulceration and torsion
- With/without diarrhea



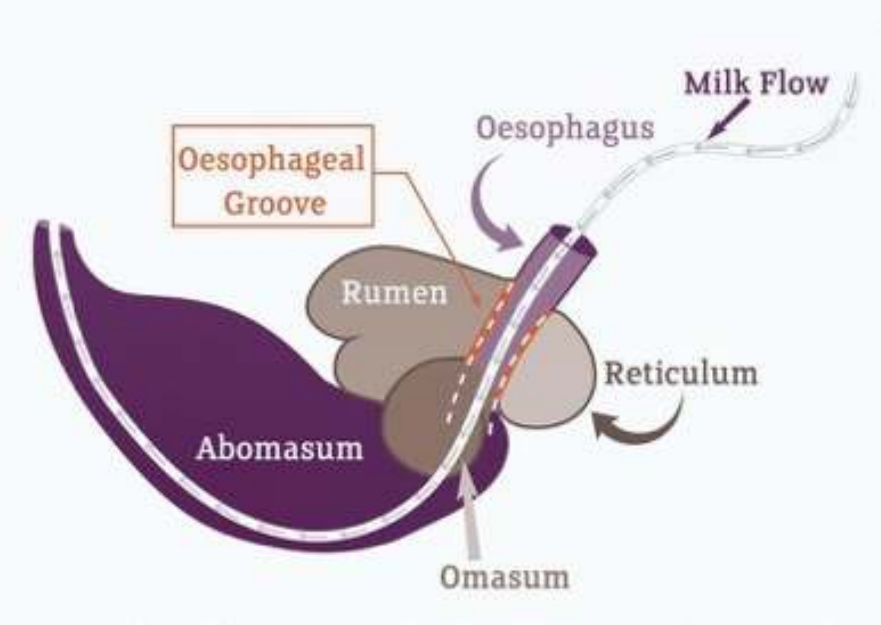
(Mee, 2023)

Abomasal syndrome

- Predisposing factor: Management
  - delayed abomasal emptying: feeding a large volume (>3 L) / high osmolality (>15% total solids) of liquid
    - highly fermentable carbohydrate → excessive gas → reduced abomasal motility
  - poor hygiene
  - Inadequate water temperature

Abomasal syndrome

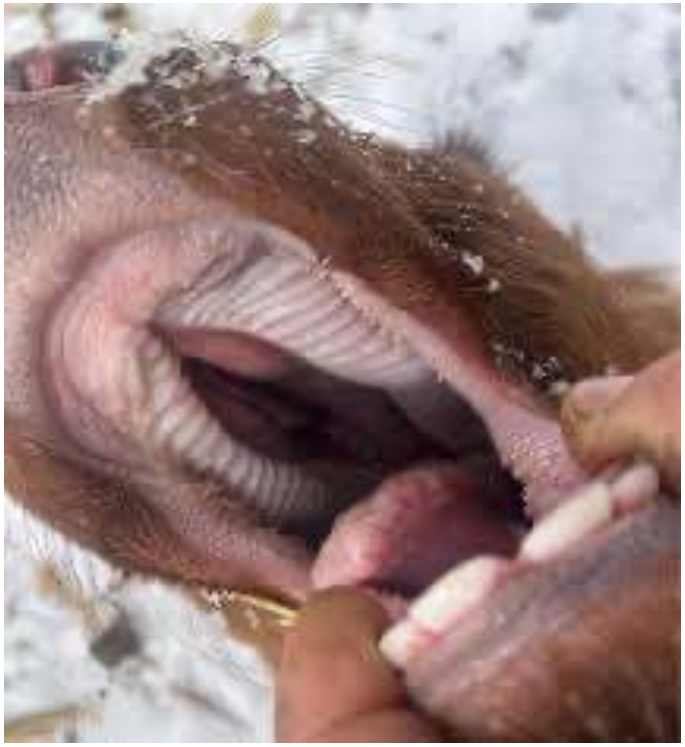
- Prevention:
  - good hygiene of the feeding materials
    (machine, bottle, nipple, bucket)
  - appropriate water temperature for milk replacer
  - Bottle feeding: forming esophageal groove



https://reurl.cc/jVaAYy

Congenital defect

- Cleft palate:
  - the roof of the mouth (hard and/or soft palate) fails to close completely during gestation, leaving an opening between the oral and nasal cavities
  - Milk coming from the nose
    poor growth / body condition
    Aspiration pneumonia
  - Palatoplasty



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Congenital defect

- Flexural deformities:
  - fetlock/metacarpophalangeal joint or carpal joint
  - Tendons responsible for limb flexion are shorter than the bones



Congenital defect

- Flexural deformities treatment:
  - Mild: standing (stretching the tendons)
  - Moderate: oxytetracycline (6–10 mg/kg) dilute in saline, IV
    splint and bandage
    standing
  - Severe: tenotomy
    splint and bandage
    standing